

24 Alternating Current

For resistive load
 $\langle P \rangle = P_0/2$

$$I = I_0 \sin \omega t$$

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

period,
frequency,
peak value
and root-mean
square value

24.1 Characteristics of alternating current

Distinguish
between peak and
rms value ...

24. Alternating Currents

24.2 Transformer

Principle of
operation

Half-wave and
full-wave

24.4 Rectification

effect of single
capacitance on
smoothing

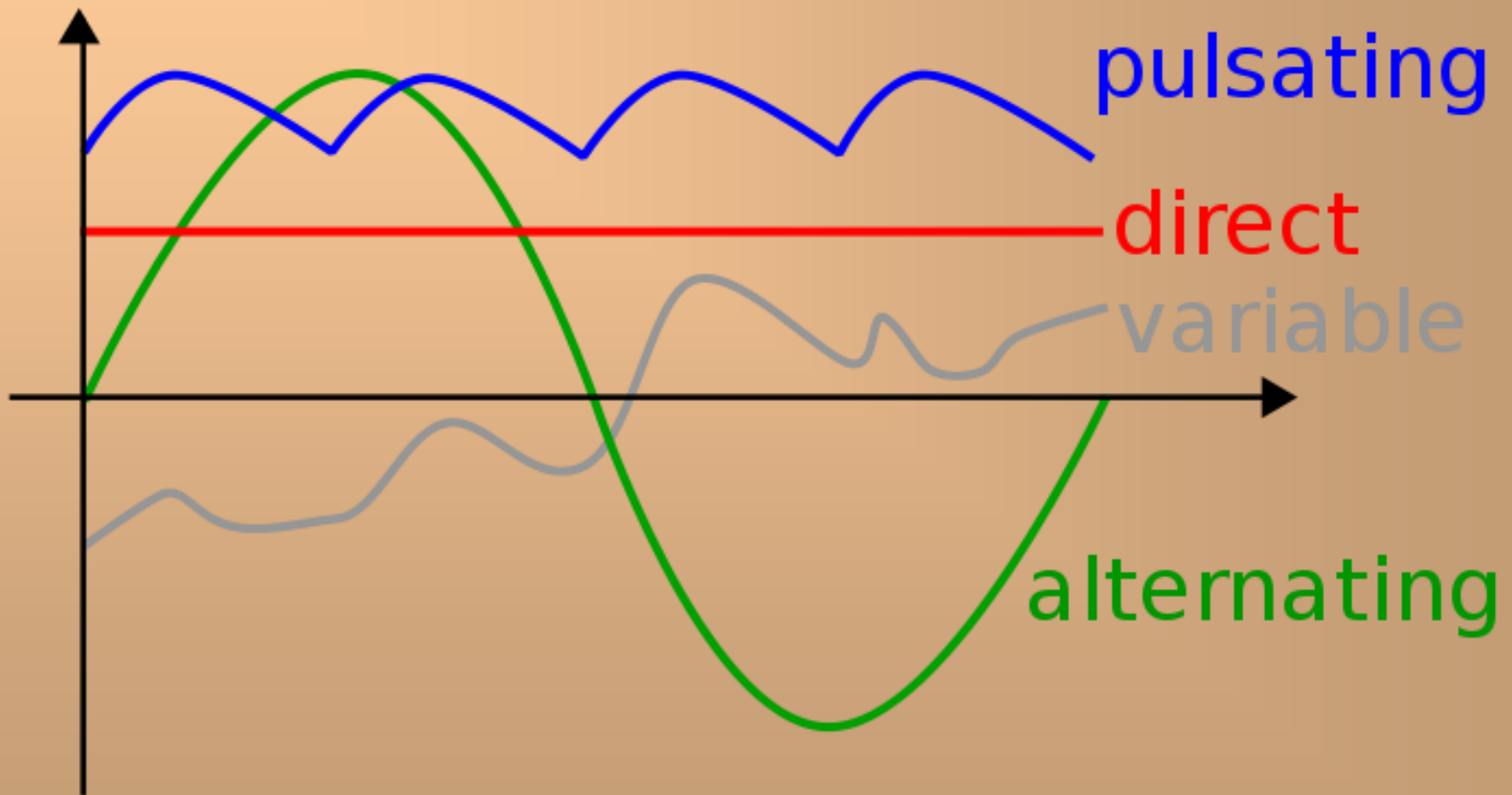
Single diode and
bridge rectifier

24.3 Transmission of electrical energy

Advantage of ac
and high voltages

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}$$

Terms you are likely to encounter



Alternating Current (AC) Electricity

by Ron Kurtus (revised 2 June 2009)

Alternating current (AC) electricity is the type of electricity commonly used in homes and businesses throughout the world. While direct current (DC) electricity flows in one direction through a wire, AC electricity alternates its direction in a back-and-forth motion. The direction alternates between 50 and 60 times per second, depending on the electrical system of the country.

<http://www.school-for-champions.com/science/ac.htm>

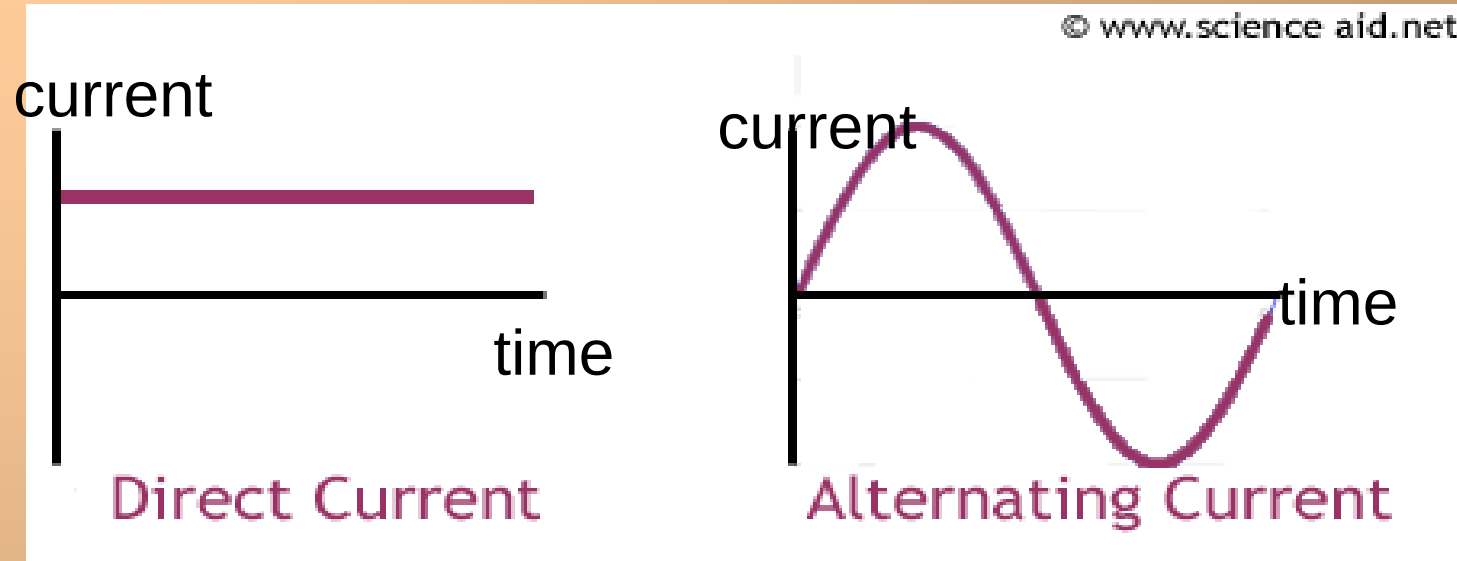
Alternating Current (AC) Electricity

by Ron Kurtus (revised 2 June 2009)

AC electricity is created by an AC electric generator, which determines the frequency. What is special about AC electricity is that the voltage can be readily changed, thus making it more suitable for long-distance transmission than DC electricity. But also, AC can employ capacitors and inductors in electronic circuitry, allowing for a wide range of applications.

<http://www.school-for-champions.com/science/ac.htm>

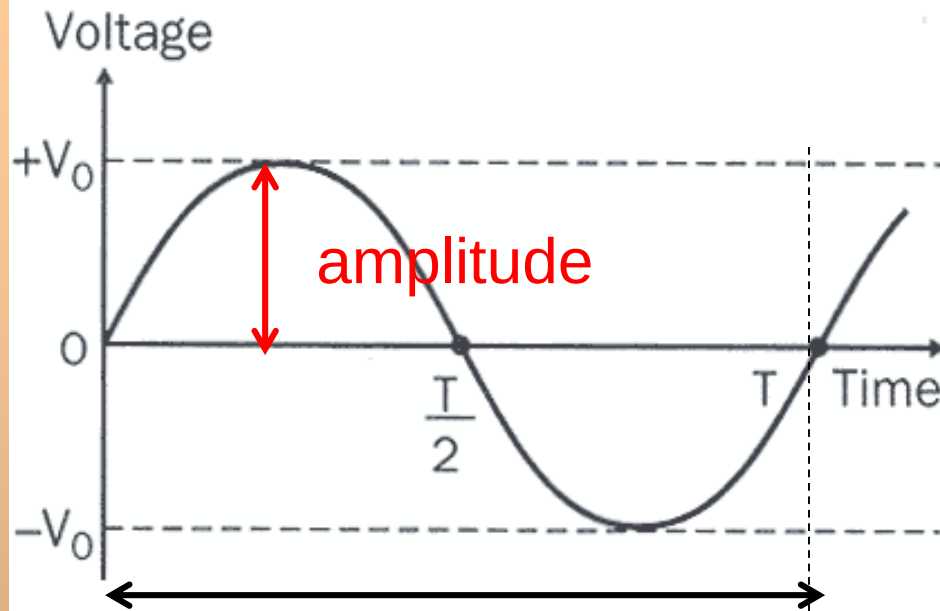
24.1 Characteristics of a.c.



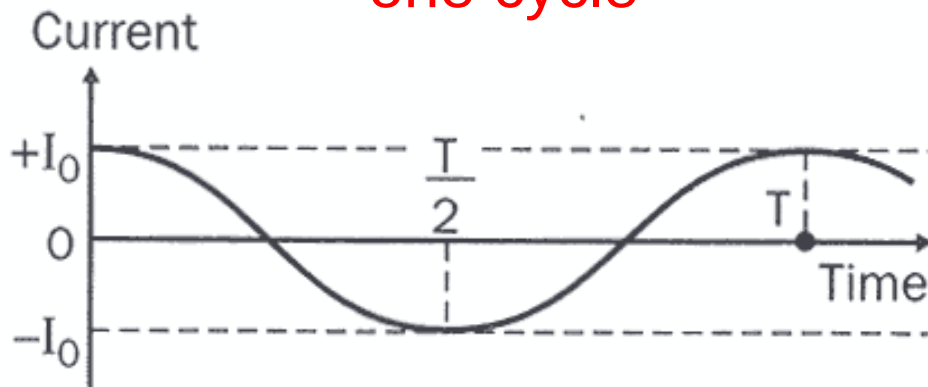
Current flow in
one direction.

An ***alternating current*** (a.c.) is an electric current that periodically reverses its direction in the circuit, with a frequency f independent of the constants of the circuit.

Terms



one cycle



Current versus time.

- a) Peak or maximum current (I_0 , or I_m) is the **maximum** current or amplitude of current.
- b) One cycle is one **alternation**.
- c) Frequency (f) is the **number** of cycles occurring per second.
- d) Period (T) is the **time** for one complete cycle.

$$T = 1/f$$

Equations of graph?

Equation of sinusoidal alternating current

a) The instantaneous current is the current at any time t is given by,

$$\mathbf{i = I_0 \sin (\omega t + \phi)}$$

where I_0 = peak or maximum current

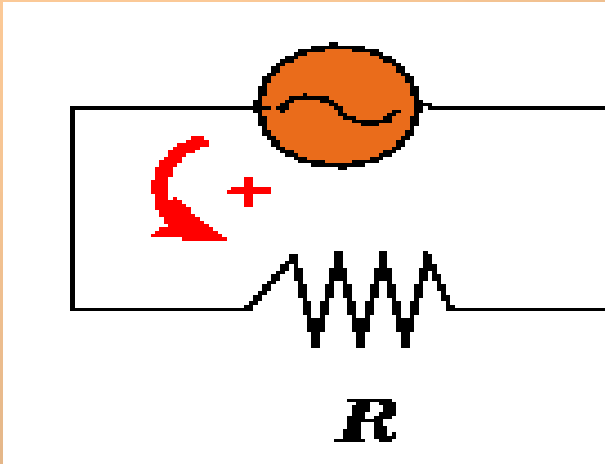
t = time

$\omega = 2\pi f$ = angular frequency

ϕ = phase

f = frequency of a.c.

A.C. flowing in a resistor,



The voltage from the source
is

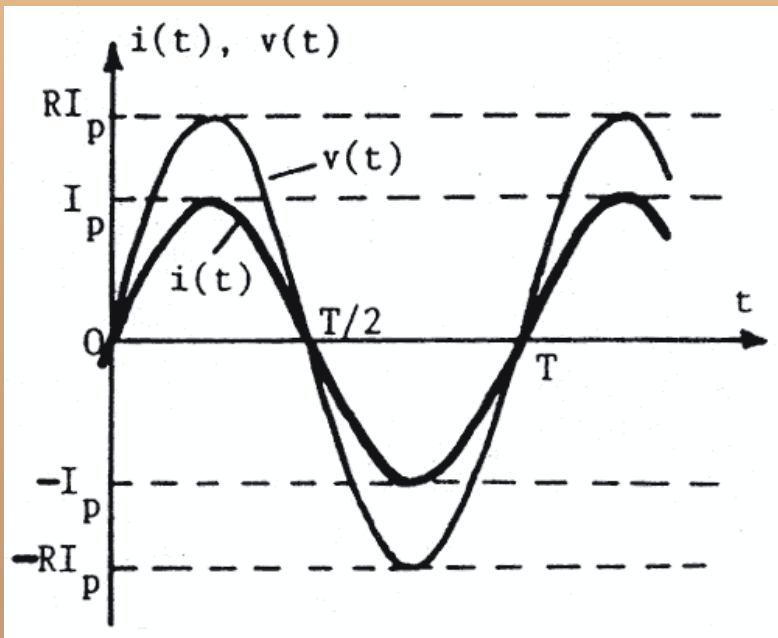
$$v = V_o \sin \omega t$$

Current through the load,

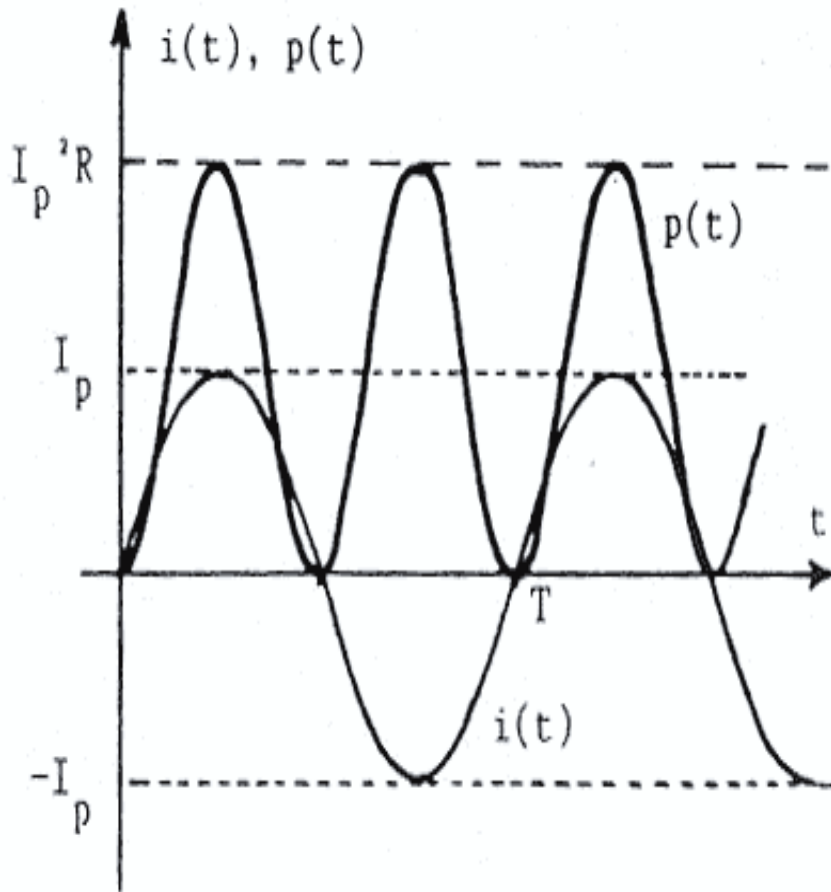
$$i = \frac{v}{R} = I_o \sin \omega t$$

Where $I = V_o / R$

The current and voltage are
in phase.



Power



The instantaneous power developed across the resistor is

$$P = i^2 R$$

$$= P_0 \sin^2 \omega t$$

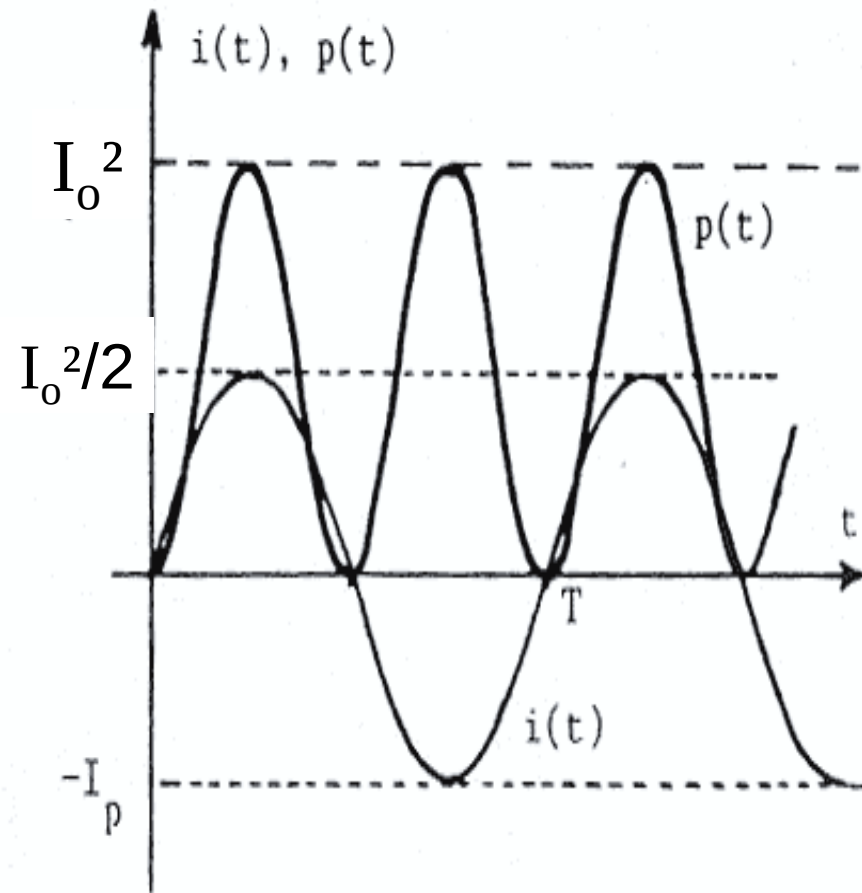
where $P_0 = I_0^2 R$

= peak or maximum power

<http://www.ngsir.netfirms.com/englishhtm/Rms.htm>

<http://www.phys.unsw.edu.au/PHYS1111/acc/acc.html>

Mean power

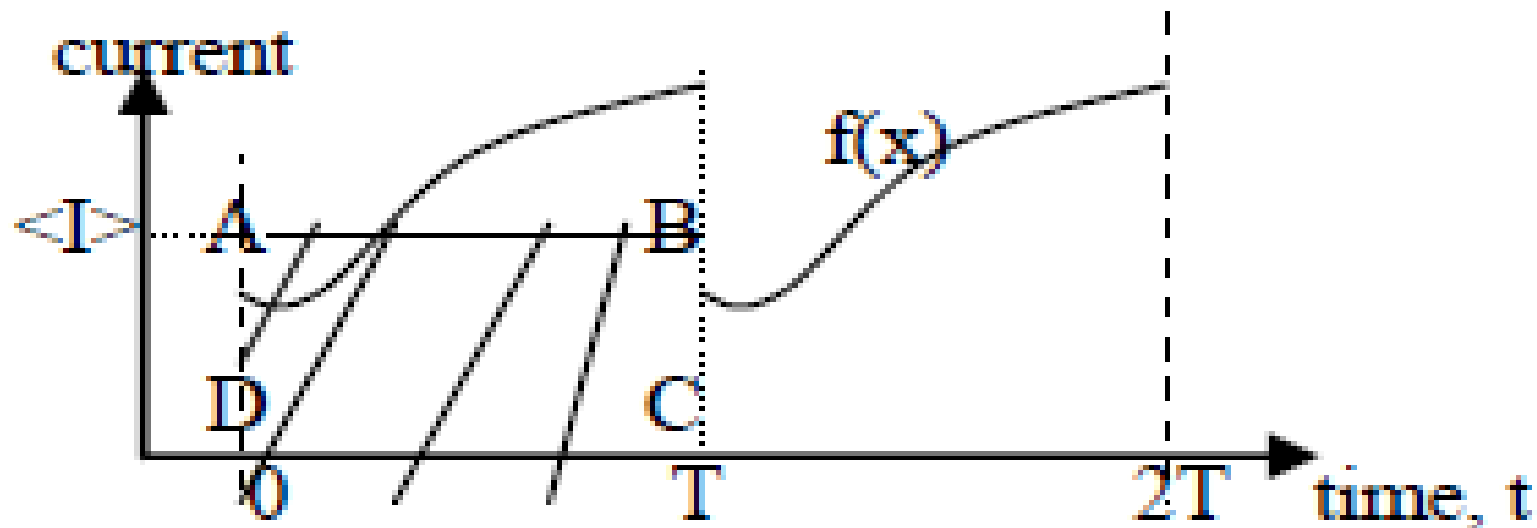


The equation $i^2 = I_o^2 \sin^2 \omega t$ is also a sinusoidal curve with a mean value $I_o^2/2$.

$$\begin{aligned} \langle P \rangle &= \frac{R}{T} \int_0^T I_o^2 \sin^2 \omega t \, dt \\ &= \frac{R I_o^2}{2} \\ &= \frac{P_o}{2} \end{aligned}$$

The mean power is equal to half the maximum power.

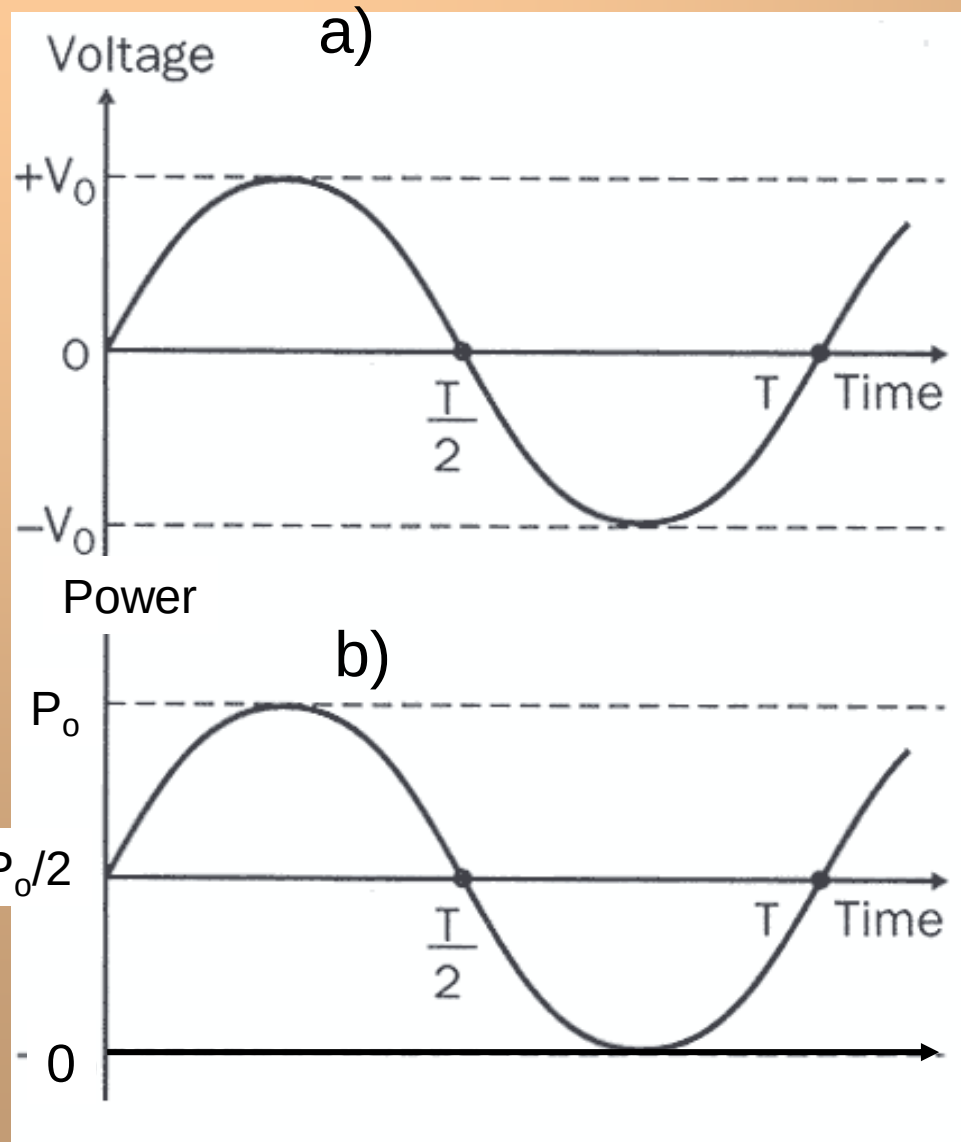
Review: Mean value



area of rectangular ABCD = area under the graph

$$\langle I \rangle \cdot T = \int_0^T f(t)$$

Exercise 24.2



Determine the mean of the curves shown.

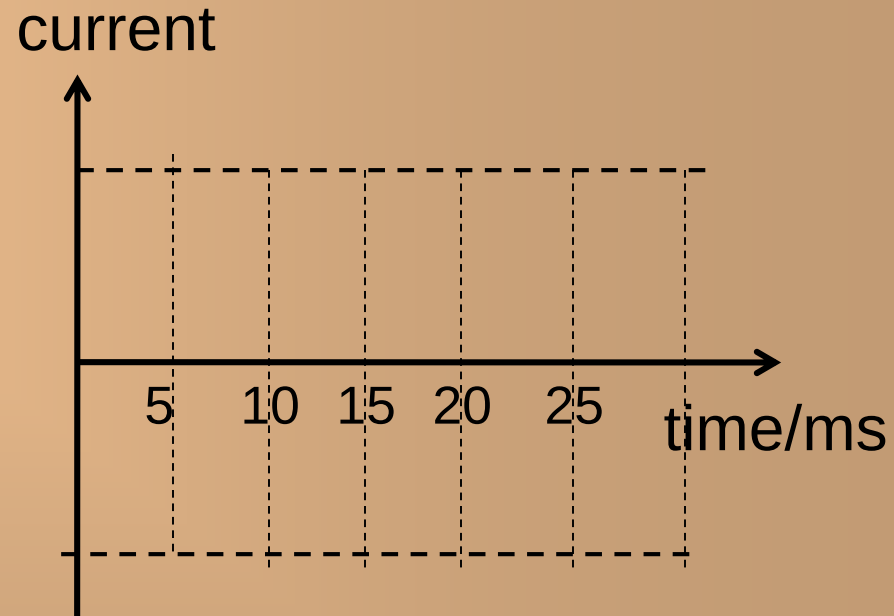
Soln

a) $\langle V \rangle = 0$

b) $\langle P \rangle = P_0/2$

Exercise 24.3

Sketch a graph of an alternating current of amplitude 20 A, 50 Hz and a phase of $\pi/2$ radian. **What is the equation of the graph?**



Pulse generator [oscillator]



Provides alternating
current of different
frequency and
amplitude.





SIGNAL GENERATOR



symbol

- Other names:
Oscillator,
- low voltage a.c., pulse generator
- Provides different waveforms in the form of a.c.
- Allow different frequencies and amplitudes of sinusoidal wave forms to be selected.
- Variation in voltage.

Root mean square current.

Mean Power

$$\langle P \rangle = \frac{I_o^2}{2} R$$

$$\langle P \rangle = I_{rms}^2 R$$

where

I_{rms} = root mean square
current

For d.c., $P = I^2 R$

$$I_{rms} = \frac{I_o}{\sqrt{2}}$$

The root mean square [rms] current is the a.c. that has the same heating effect in a given resistor as a direct current (d.c.)

Mathematics corner [for info]

$$\langle I^2 \rangle = \frac{1}{T} \int_0^T I_o^2 \sin^2 \omega t \cdot dt$$

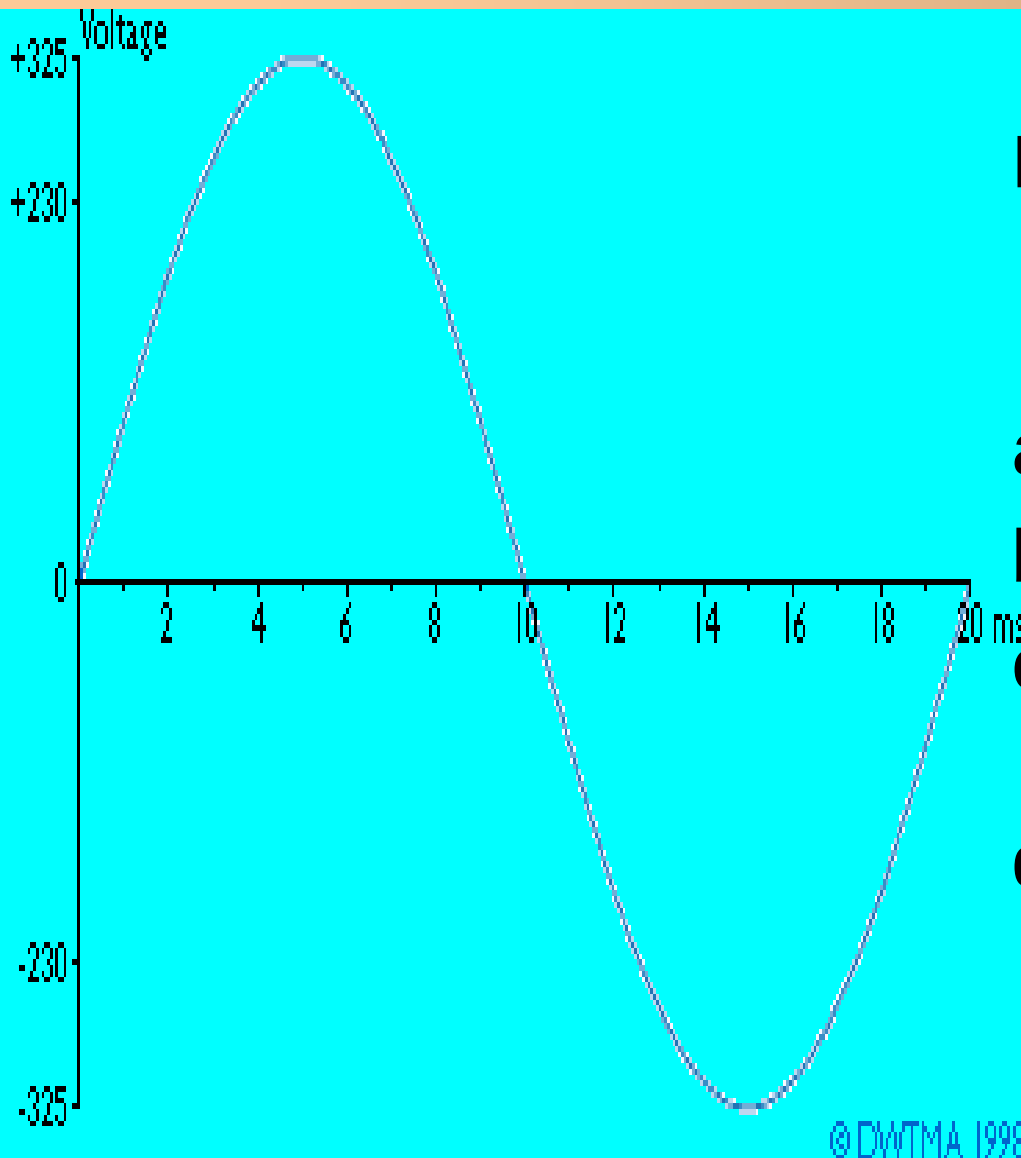
$$= \frac{I_o^2}{2T} \int_0^T (1 - \cos 2\omega t) dt$$

$$= \frac{I_o^2}{2T} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T$$

$$= \frac{I_o^2}{2} \text{ as } \omega = \frac{2\pi}{T}$$

$$\begin{aligned} \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= 1 - 2 \sin^2 \theta \end{aligned}$$

Exercise 24.4



For the alternating voltage shown, determine,

- a) peak voltage,
- b) frequency,
- c) root mean square voltage,
- d) state the equation of the sinusoidal voltage.

Exercise 24.5

$$i = I_o \sin \omega t$$

$$I_{rms} = \frac{I_o}{\sqrt{2}}$$

An alternating current of

$$i = [5.0/\text{A}] \sin [100 \text{ t/s}]$$

passes through a load of resistance 20Ω . The current is measured in ampere and the time in second. Calculate,

- a) the peak current, **5.0 A**
- b) the root mean square current,
- c) the frequency of the current,
- d) the period,

$$I_{rms} = \frac{5}{\sqrt{2}} = 3.54 \text{ A}$$

$$\omega = 100, \omega = 2\pi f$$
$$f = 15.9 \text{ Hz}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{100}$$
$$= 62.8 \text{ ms}$$

Exercise 24.5

An alternating current of
 $i = 5.0 \sin [100 t/\text{s}]$

Calculate,

- e) the instantaneous current at 70 ms,
- f) the maximum power dissipated
- g) the mean power dissipated,

(Ans. a) 5.0 A, b) 3.54 A ,
c) 15.9 Hz, d) 62.8 ms
e) 3.2 A, f) 500 W
g) 250 W)

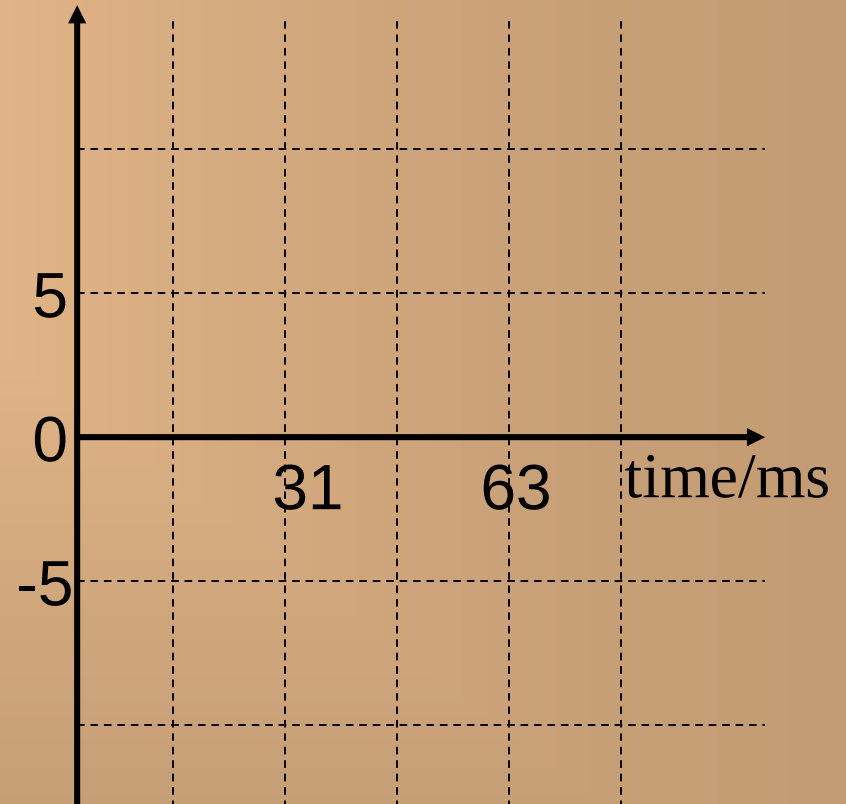
$$\begin{aligned} \text{e) } I &= 5 \sin 100(0.07) \\ &= 3.3 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{f) } P_o &= I_o^2 R = 5^2 20 \\ &= 500 \text{ W} \end{aligned}$$

$$\text{g) } \langle P \rangle = 500/2 = 250 \text{ W}$$

Exercise 24.5

- a) Sketch the graph of i versus time.
- b) Sketch the graph of the instantaneous power versus time.
- c) Sketch the graph of double the peak current but half the frequency.



How to calculate root mean square current.

Start from the back of the phrase.

a) Find the sum of square of the quantity,

$$a) \quad \sum_{i=1}^N I_i^2 = I_1^2 + I_2^2 + \dots + I_N^2$$

b) Find the mean of a), and

$$b) \quad \langle I^2 \rangle = \frac{\sum I^2}{N}$$

c) Take the square root of b),

$$c) \quad I_{rms} = \sqrt{\langle I^2 \rangle}$$

Exercise 24.6

Find a) the mean voltage,
and b) the mean
square voltage and c)
root mean square
voltage.

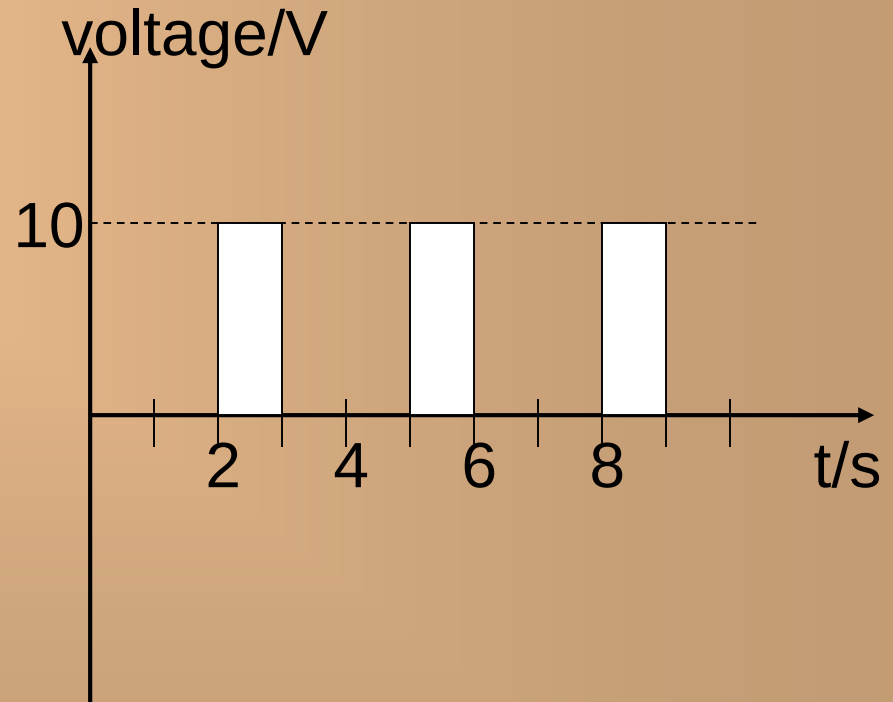
(Ans. a) 3.33 V b) 33.3 V²,
c) 5.77 V)

Soln.

a) $3\langle V \rangle = 10[1]$
 $\langle V \rangle = 3.33 \text{ V}$

b) $3\langle V^2 \rangle = 100[1]$
 $\langle V^2 \rangle = 33.3 \text{ V}^2$

c) $V_{\text{rms}} = \sqrt{33.3} = 5.77 \text{ V}$



Self-test 24.1

- 1) What is an alternating current?
- 2) What is the peak voltage?

- 1) An ***alternating current*** (a.c.) is an electric current that periodically reverses its direction in the circuit, with a frequency f independent of the constants of the circuit.
- 2) Peak or maximum voltage is the maximum voltage or amplitude of voltage.

Self-test 24.1

3) Distinguish between the peak and root mean square voltage.

3) Peak voltage is the maximum voltage while root mean square voltage *is the alternating voltage that has the same heating effect in a given resistor as a direct voltage. The peak voltage is larger than rms voltage.*

Self-test 24.1

- 4) What is meant by
- a) frequency,
 - b) the root mean square current of an a.c.?

4a) Frequency (f) is the number of cycles occurring per second.

b) ***The root mean square [rms] current is the a.c. that has the same heating effect in a given resistor as a direct current (d.c.)***

Self-test 24.1

5) For an alternating voltage,

$$v = [20/\text{V}] \sin [200\pi t/\text{ms}]$$

Determine

- a) the peak voltage,
- b) the rms voltage,
- c) frequency of the alternating voltage.

5a) 20 V

b) $20/\sqrt{2} = 14.1 \text{ V}$

c) $2\pi f = 200\pi / 10^{-3}$

$f = 100 \text{ kHz}$

PYP 24.1

The magnetic flux density B of the field due to a long straight wire is given by

$$B = \frac{\mu_o I}{2\pi d}$$

An overhead power cable carries an alternating current of 2000 A r.m.s. At what distance would the peak magnetic flux density due to the current in the cable be $100 \mu\text{T}$?

[Ans.: 5.7 m]

Soln.

peak current I_o

$$= 2000\sqrt{2}$$

$$= 2828 \text{ A}$$

$$B = \frac{\mu_o I}{2\pi d}$$

$$100 \times 10^{-6} = \frac{4\pi \times 10^{-7} (2828)}{2\pi d}$$

$$d = 5.7 \text{ m}$$

4 An a.c. supply is connected to a resistor. When the peak value of the e.m.f. of the supply is V_0 and the frequency is f , the mean power dissipated in the resistor is P . The supply frequency is then changed to $2f$, the peak value of the e.m.f. remaining as V_0 .

475

What is now the mean power in the resistor?

PYP 24.2

A P B $\sqrt{2}P$ C $2P$ D $4P$

Ans A

$$P = \frac{1}{2} P_0 = \frac{V_0^2}{2R}$$

Is independent of frequency

24.2 Transformer

A ***transformer*** changes i.e. transforms an alternating p.d from one value to another of greater (step-up) or smaller value (step-down) using the mutual induction principle.

Power Transformer

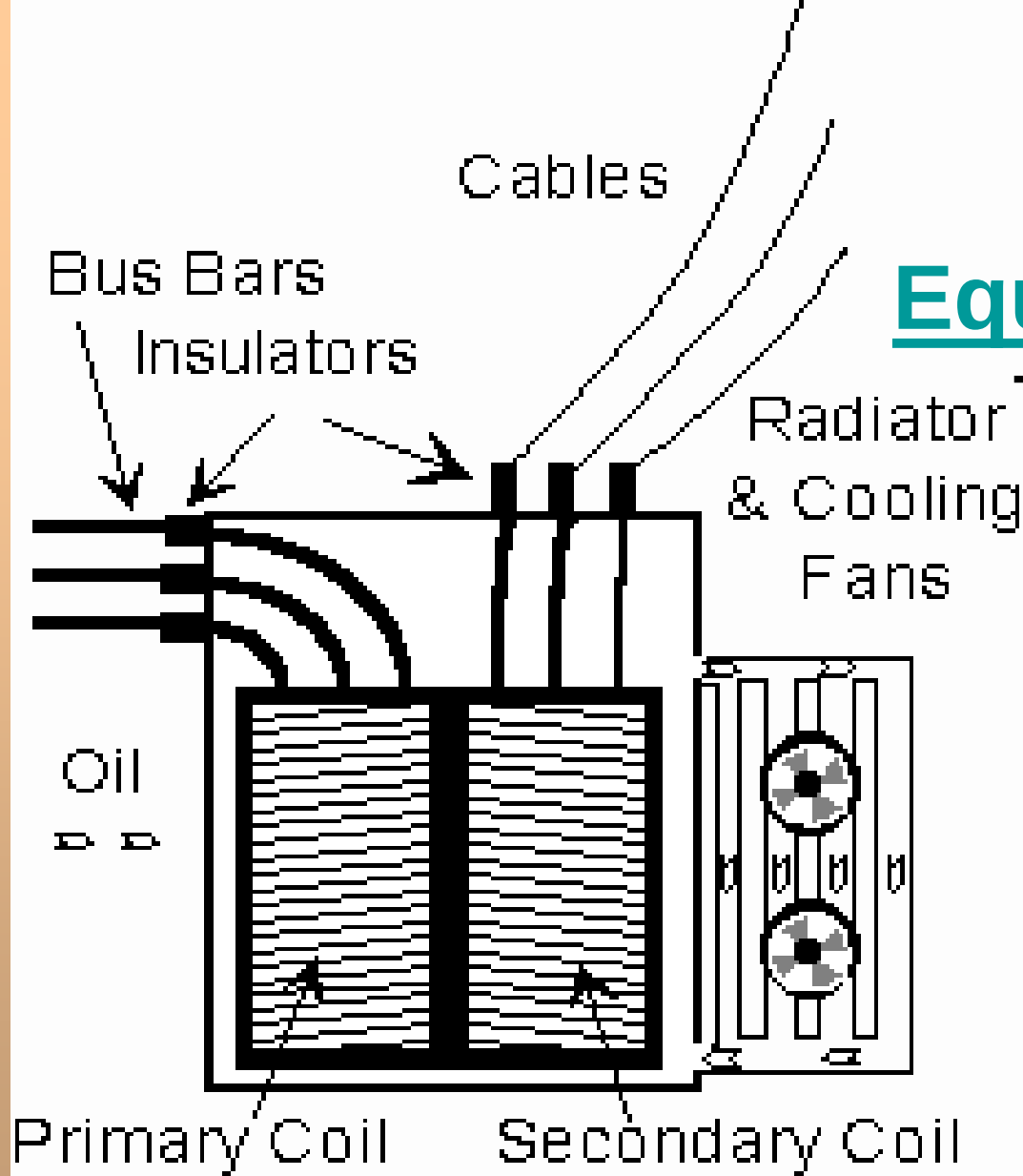
2.5 MVA General Electric Unit Substation Transformer



500 MVA Single-phase autotransformers

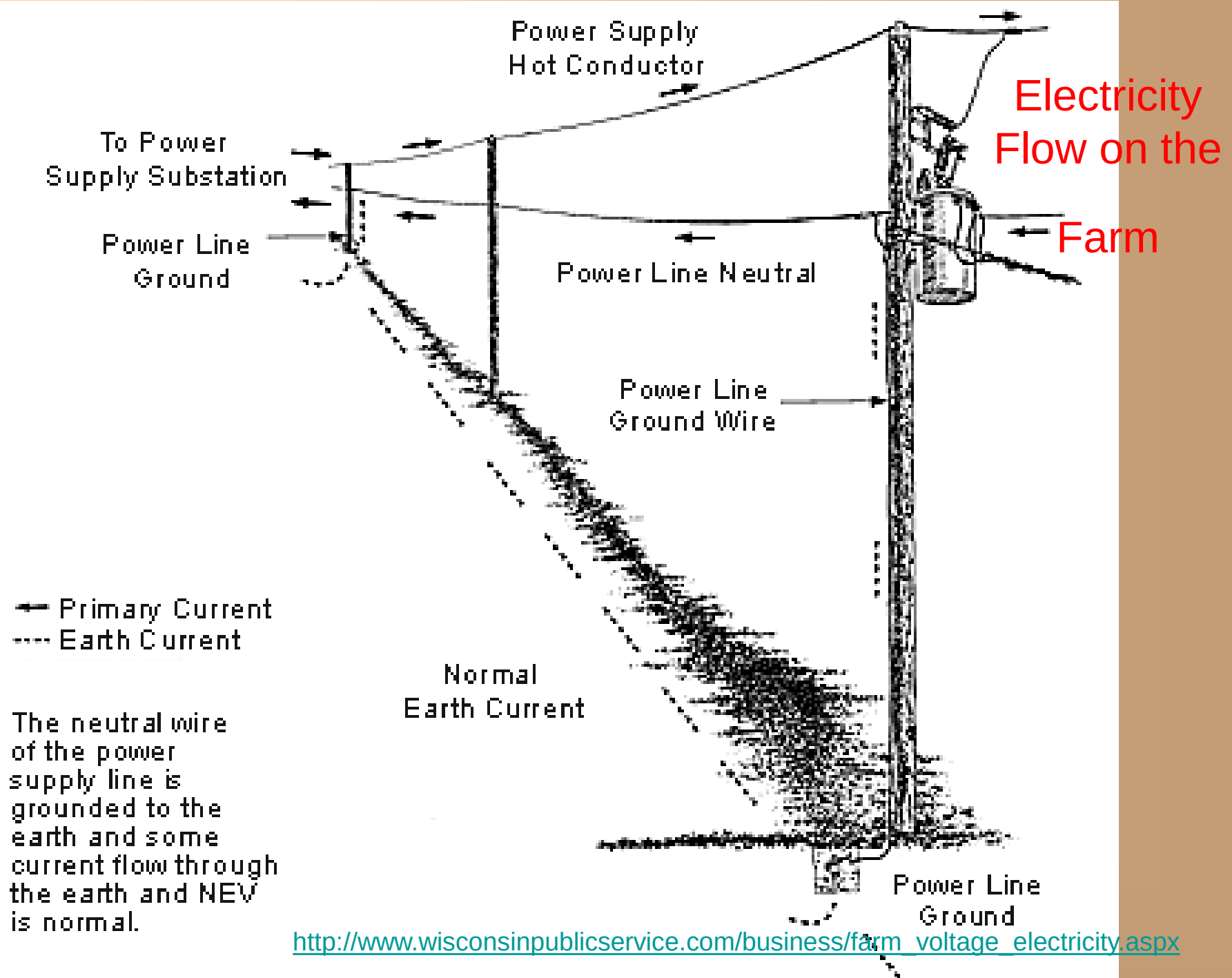


Substation Equipment: Power Transformers



STEP-UP TRANSFORMER

http://www.osha.gov/SLTC/etools/electric_power/illustrated_glossary/substation_equipment/power_transformers.html

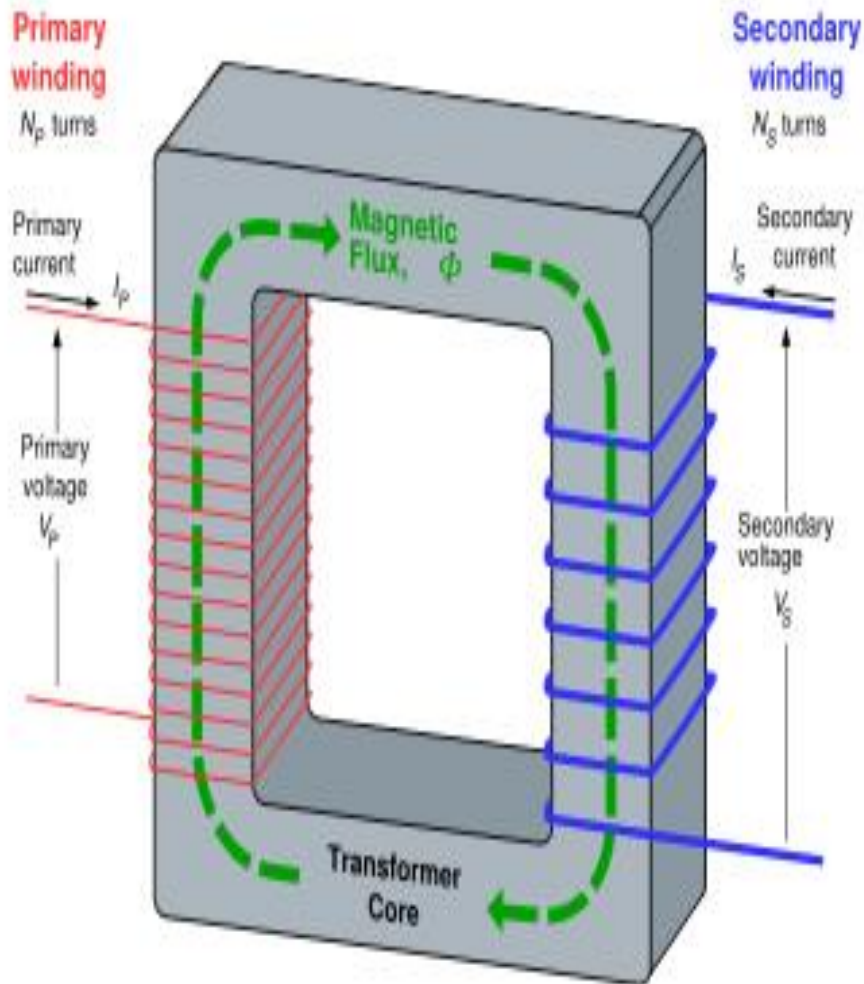


Transformer (electrical appliances)

Used in laboratory power supply.



Description

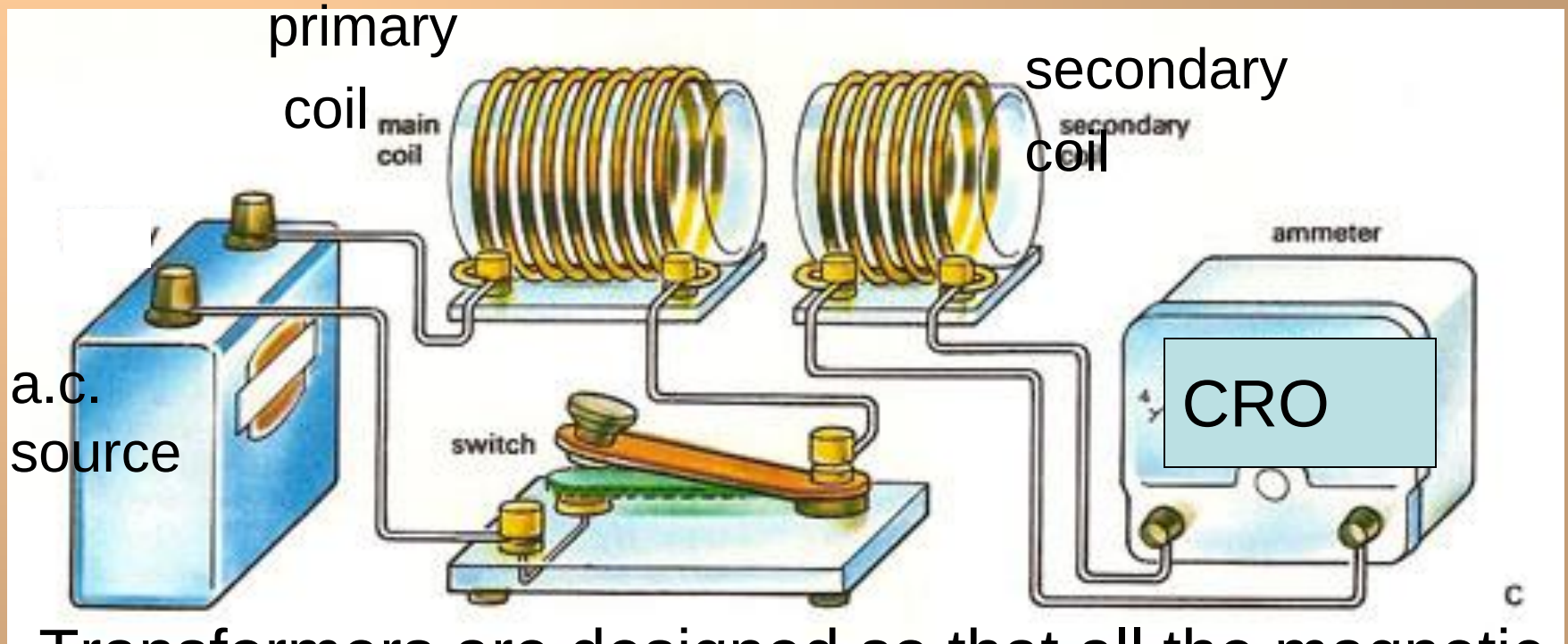


- a) A simple transformer consists of two coils, the primary and the secondary coils wound over a core made of magnetically soft material.
- b) There is no electrical connection between the primary and secondary coils, but the soft iron core provides a magnetic link between them.

Description

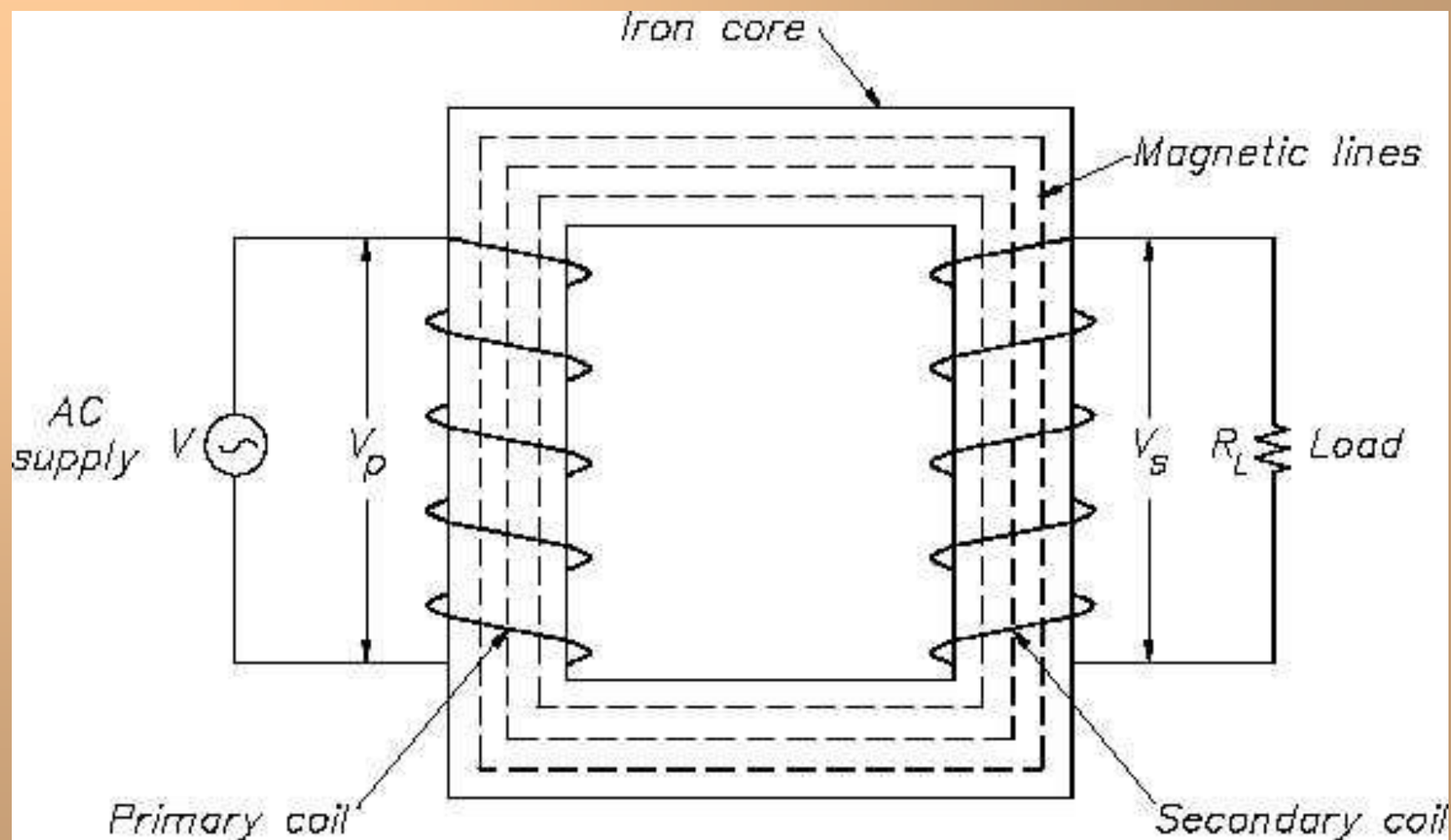
- a) An alternating voltage applied to the primary coil produces an a.c. through it, which produces an alternating magnetic flux in the core threading the secondary coil.
- b) An alternating voltage is induced in the secondary coil.
- c) Frequency of secondary voltage is the same as the primary voltage.

Transformer



Transformers are designed so that all the magnetic flux produced by the primary coil passes through the secondary. The primary coil is connected to an a.c. source.

http://www.daviddarling.info/encyclopedia/E/electromagnetic_induction.html



Principle [how it works]

When current rises in the primary coil, the magnetic field through the secondary coil due to this current increases.

The changing flux through the secondary causes e.m.f to be induced in the secondary coil

- As the current reverses direction, the emf in the secondary reverses direction.
- the frequency of the secondary is the same as the primary.

Main causes of *energy losses*

- a) **R**esistance of coil. Power dissipated in the resistor is i^2R where R is the resistance of the resistor and i the current passing through the coil. This is reduced by using low resistance thick copper wire.
- b) **E**ddy current. The changing flux in the core will cause an induced current called eddy current to flow. Laminating the core reduces the energy losses due to eddy current.

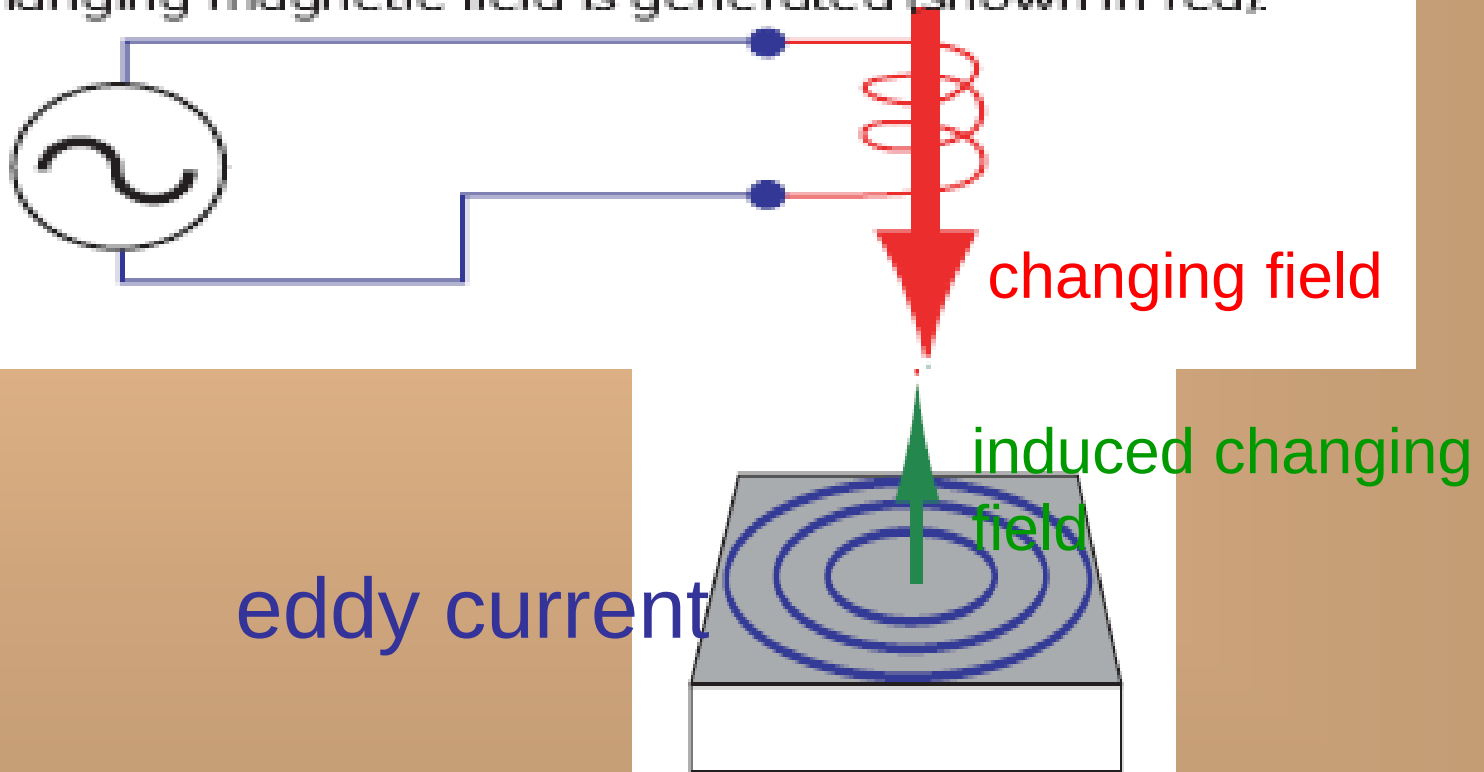
Main causes of *energy losses* [2]

- c) Flux leakage occurs when the changing flux from the primary threads the secondary. Efficient core design to ensure that all the primary flux is linked with the secondary.
- d) Hysteresis loss. Magnetization of the core is repeatedly changed from one direction another and back again. This requires energy and causes the core to get hot. This is reduced by using soft magnetic material for the core.

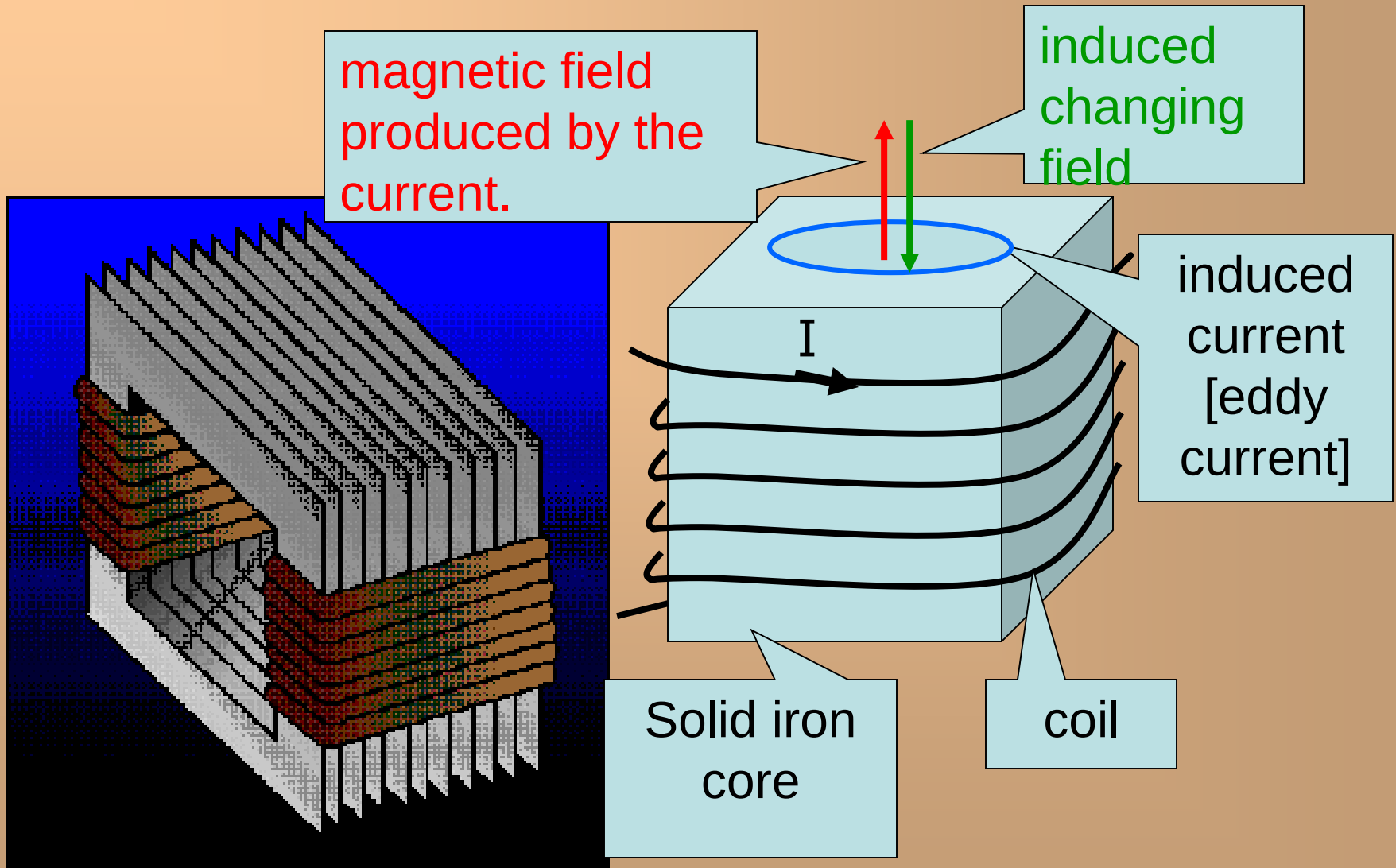
Commercial transformer has an efficiency of 95% to 99%. An ideal transformer has an efficiency of 100%.

Principles

When an alternating current is passed through a coil a changing magnetic field is generated (shown in red).



Laminated iron core



Function of *soft iron core*:

- a) concentrates the magnetic flux,
- b) laminated to reduce eddy current losses.

Lamination. The core of a transformer is formed of a pile of thin iron or steel stampings (thin sheets) called lamination. These are oxidized on the surface or lightly varnished to increase the electrical resistance from one to another.

For a transformer,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

where

V_s = secondary voltage

V_p = primary voltage

N_s = number of turns in secondary

N_p = number of turns in primary.

V_s and V_p must both be either peak voltage or both rms.

*For $N_s > N_p$, it's a **step up** transformer and

if $N_p > N_s$ it's a **step down** transformer.

For an ideal transformer

Power output = power input

$$i_s V_s = i_p V_p$$

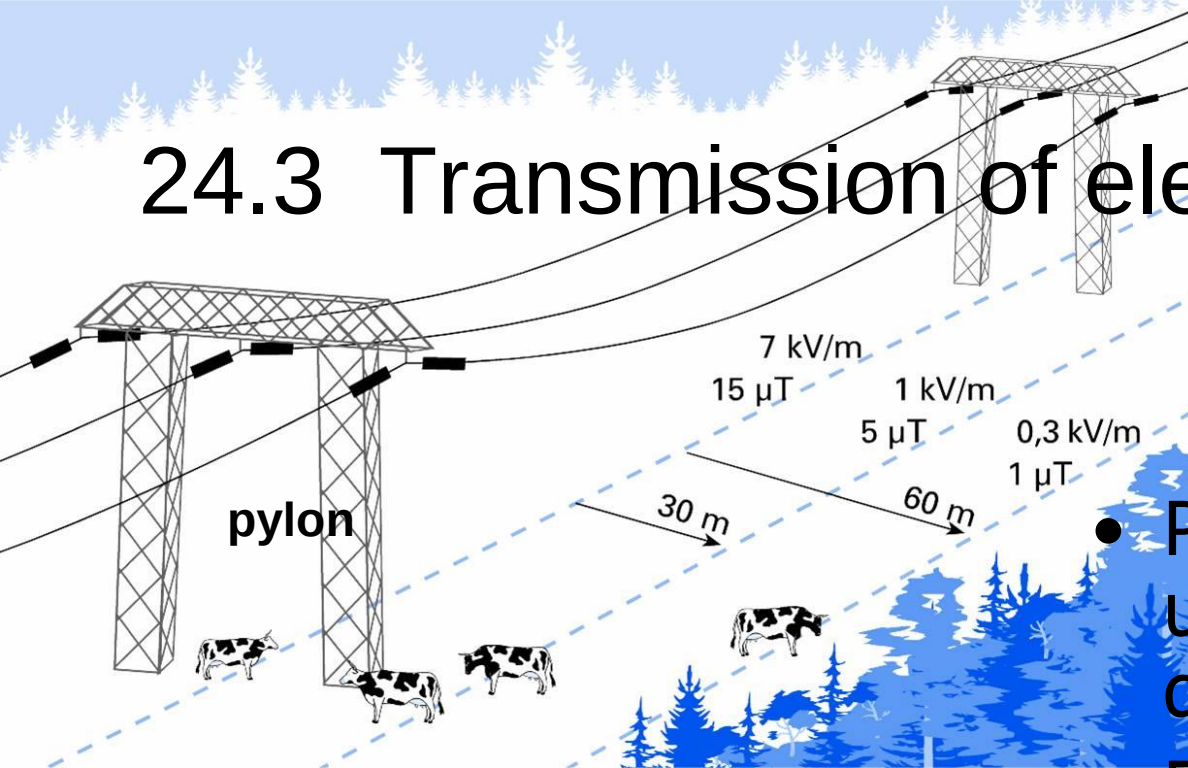
$$\frac{V_s}{V_p} = \frac{i_p}{i_s}$$

where i_p = primary current

i_s = secondary current

- For ideal transformer there is not energy lost, the input energy is completely transformed into the output energy.

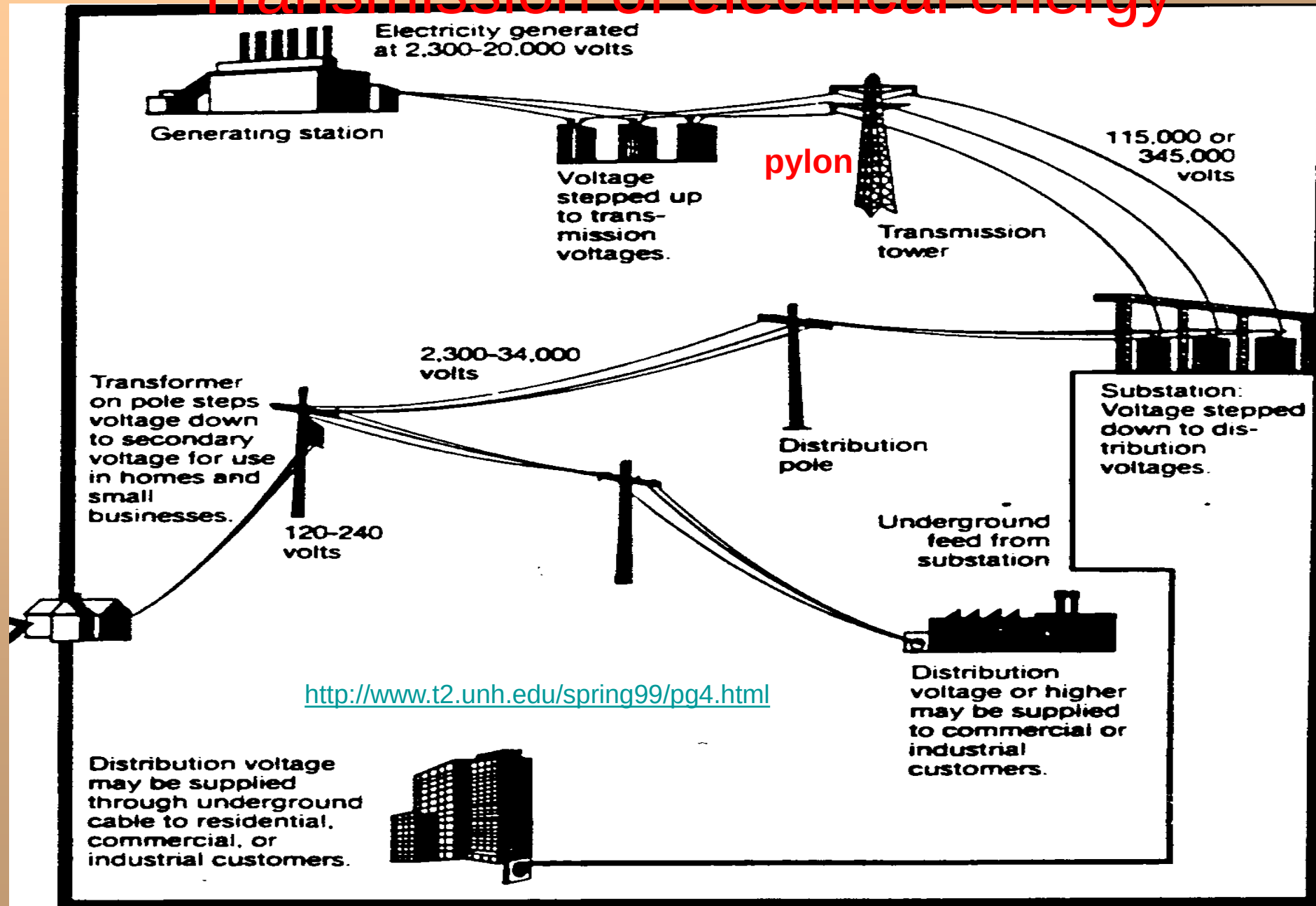
24.3 Transmission of electrical energy

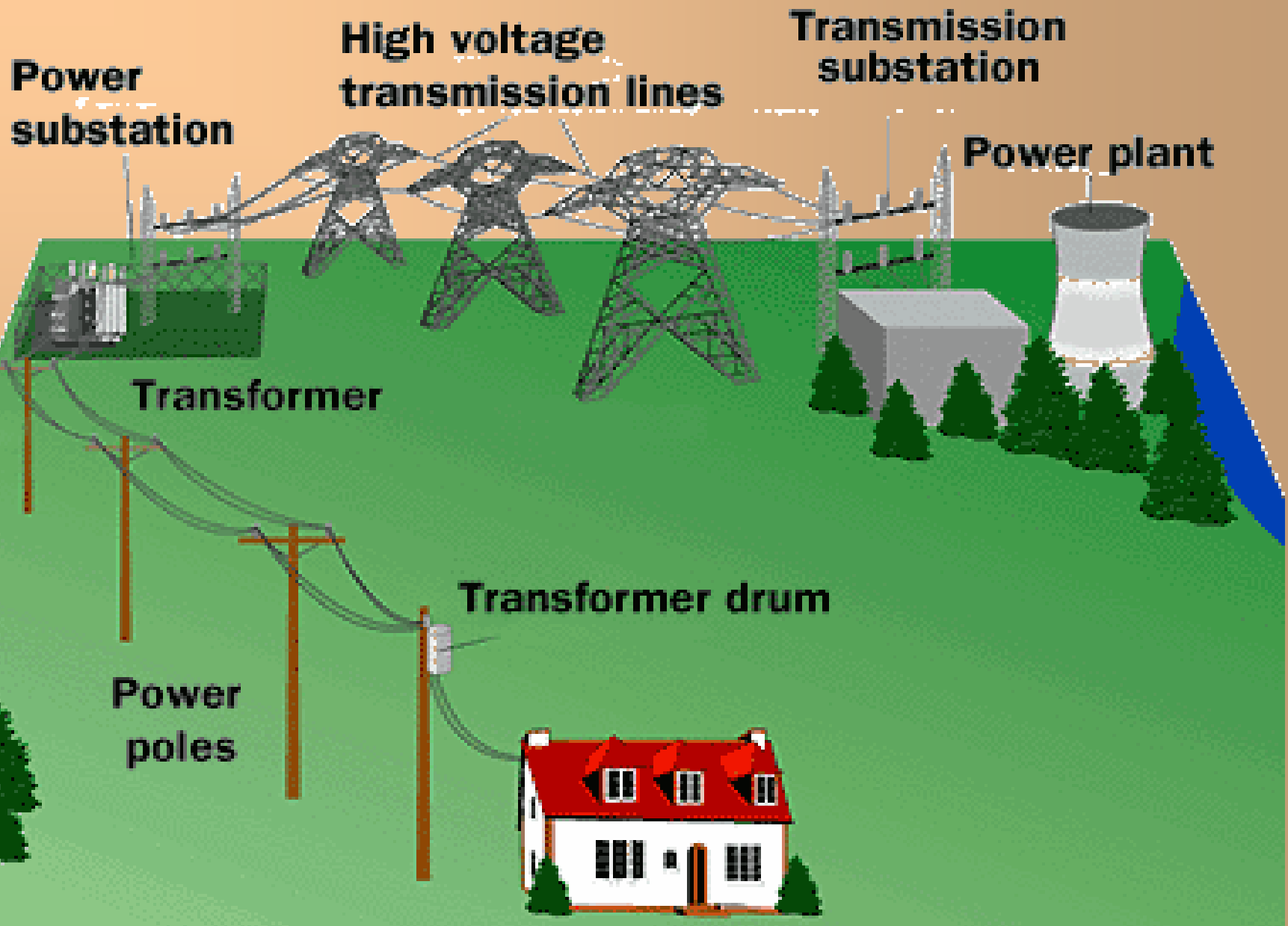


- Transformer play an essential part in the transmission of electricity.

- Power plants are usually placed some distance from towns.
- Electricity needs to be transmitted over long distance.
- There is always power loss in transmission lines due to their resistance (I^2R).

Transmission of electrical energy





Transmission

In Britain a network of cables, called the national grid, links all the power stations. It allows the demand for electricity to be shared out between the power stations. Most of the cables in the grid system are carried overhead on pylons. Underground cables are more expensive and difficult to maintain. They are used in cities and where the scenery must not be spoilt.

Advantages of using electrical energy

1. Electrical energy is the easiest form of energy to transmit, and distributed by cables.
2. For many modern appliances, electrical energy is the only form of energy that can be used.
3. Electrical energy can be converted efficiently into any one of the other forms of energy.

How electrical energy is transmitted?

Electricity is sent over long distance using cables. Transmission is done using alternating current at high voltages to reduces energy losses in cables.

1. The voltage is step up to high voltage before transmitted from power station.
2. This ensures that the current flowing in the cables is small and the rate of power dissipated in cables are minimum (I^2R).

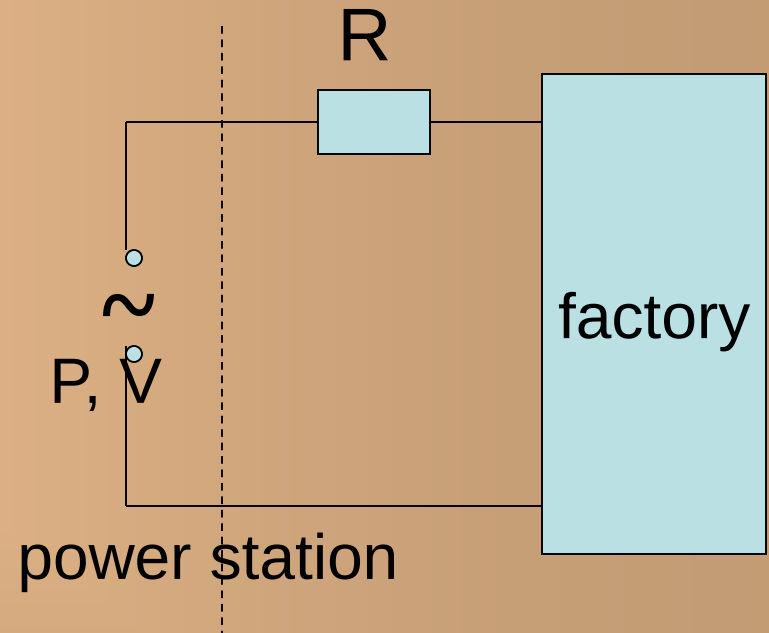
How electrical energy is transmitted?

3. Through the national grid, the voltage is lowered in stages at receiving substations depending on the need of the customer.
4. The national grid is made up of close network of cables that join receiving substations

Example 24.7

The output power P and output voltage V from a power station is connected to a factory by cables of total resistance R . Calculate

- a) the current flowing in the circuit,
- b) the power dissipated in cables,
- c) the power input to the factory.



Advantages of a.c. in transmission

1. Direct current are less easy to generate than alternating currents.
2. Alternating e.m.fs are more convenient to step up and to step down.
3. Alternating current is just as suitable for heating as direct current. The heating effect does not depend on direction of current. e.g. (a) lighting: filament lamps depend on the heating effect, gas discharge lamps run as well on alternating current. (b) small motors in vacuum cleaners can use a.c.

Advantages of a.c. in transmission [2]

4. Transmission using alternating current is more efficient than d.c. transmission. This is because high voltage transmission is more efficient than low voltage transmission.
5. In high voltage and low current transmission of electrical power, low currents require thinner and therefore cheaper cables.

Disadvantage.

For use of high voltage the high cost of the substation insulation needed. Cost of transmitting a.c. is lower than direct current.

Practical transmission system

The energy loss in the cables can be reduced in two basic ways:

- (a) By reducing the resistance of the cables.
- (b) By reducing the current flowing.

Large reductions in the resistance of the cables can only be brought about by making the cables very thick. This is not practical for several reasons.

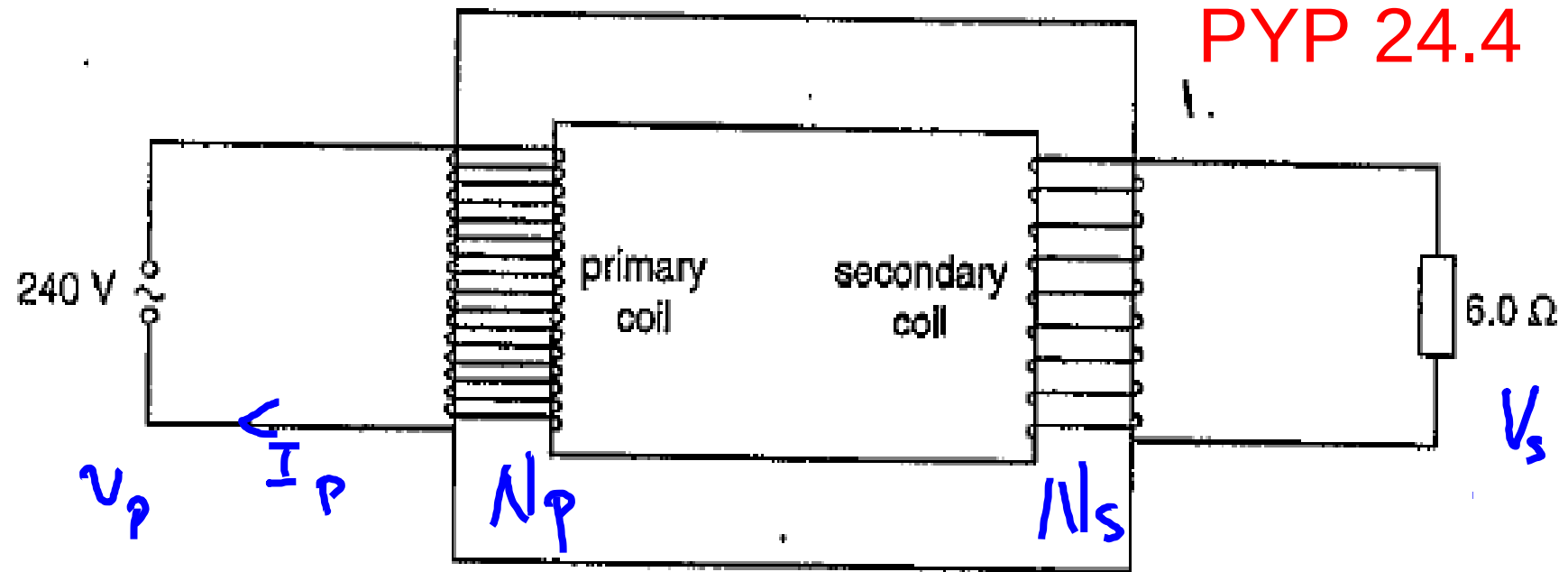
Practical transmission system [2]

1. Thicker cables are more expensive as more materials are required, expensive to manufacture and installed.
2. Thicker cables may not be slung from pylons.
3. It is more difficult and costly to insulate high voltage cables than to be laid underground.

- 1 The diagram shows an iron-cored transformer assumed to be 100% efficient. The ratio of the secondary turns to the primary turns is 1:20.

97 s

PYP 24.4



A 240 V, a.c. supply is connected to the primary coil and a 6.0Ω resistor is connected to the secondary coil.

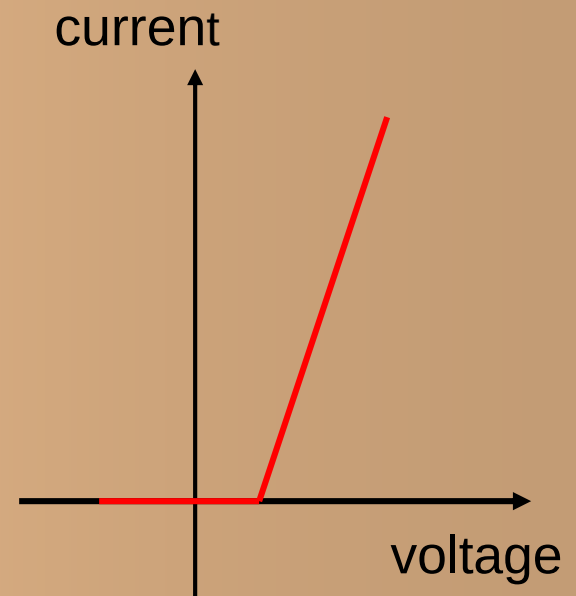
What is the current in the primary coil?

- A 0.10 A B 0.14 A C 2.0 A D 40 A

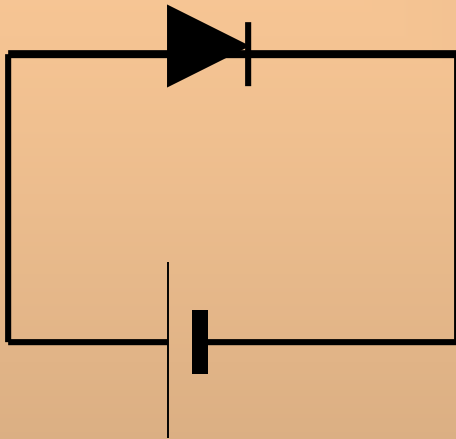
Ans: C

24.4 Rectification

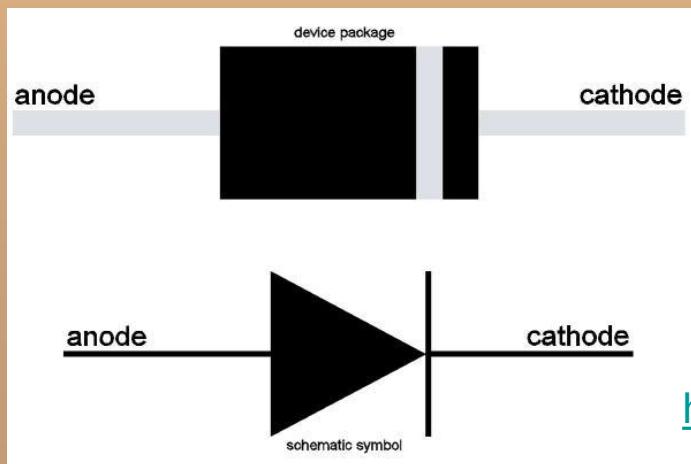
1. Rectification is the **conversion** of alternating current (a.c.) to direct current (d.c.)
2. A **rectifier** is a conductor which is largely **unidirectional**.
3. Ideal rectifier or diode.
 - a) Must have a **zero resistance** when the current flow in one direction and
 - b) Must have an **infinite resistance** when the current flows in opposite direction.



Forward biased

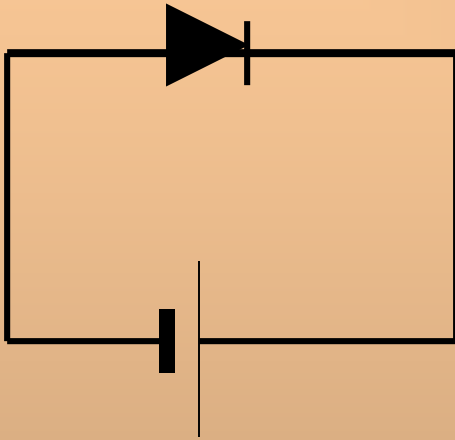


- Direction of **conventional current** is the same as direction of arrow of diode.
- Rectifier **conducts** and has a zero resistance.
- A real diode has **low** resistance



http://www.gadgetiq.com/tach_install.htm

Reversed biased

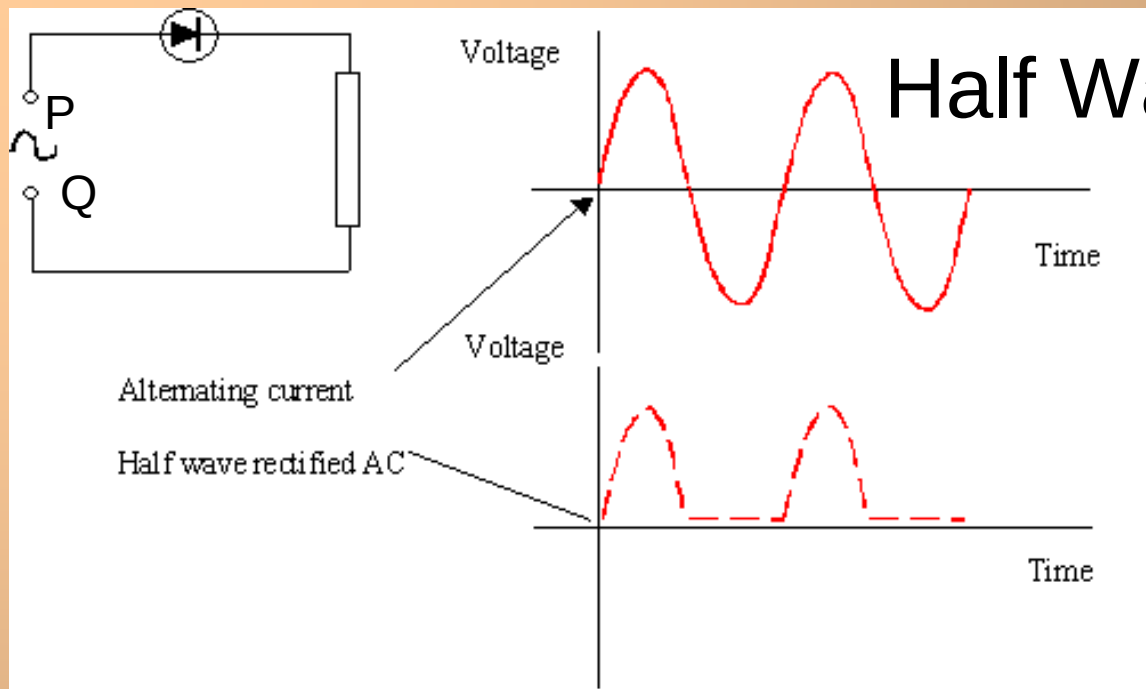


- Direction of conventional current is **opposite** to that of the arrow of the diode.
- The diode is **non-conducting**, and has an infinite resistance.
- A real diode has a **high resistance** and negligible current flows.

Why do we need to rectify a.c.?

D.C is required for

- a) battery charging
- b) operating of CRO
- c) operation of GM tube.
- d) operation of X-ray tube
- e) operation of radio receivers and transmitters.



Half Wave Rectification

During the first half cycle when P is positive, the diode is.....**forward**..... biased and is **conducting**.....

During the second half cycle when Q is negative, the diode is **reversed**..... biased, and is **non-conducting**.....

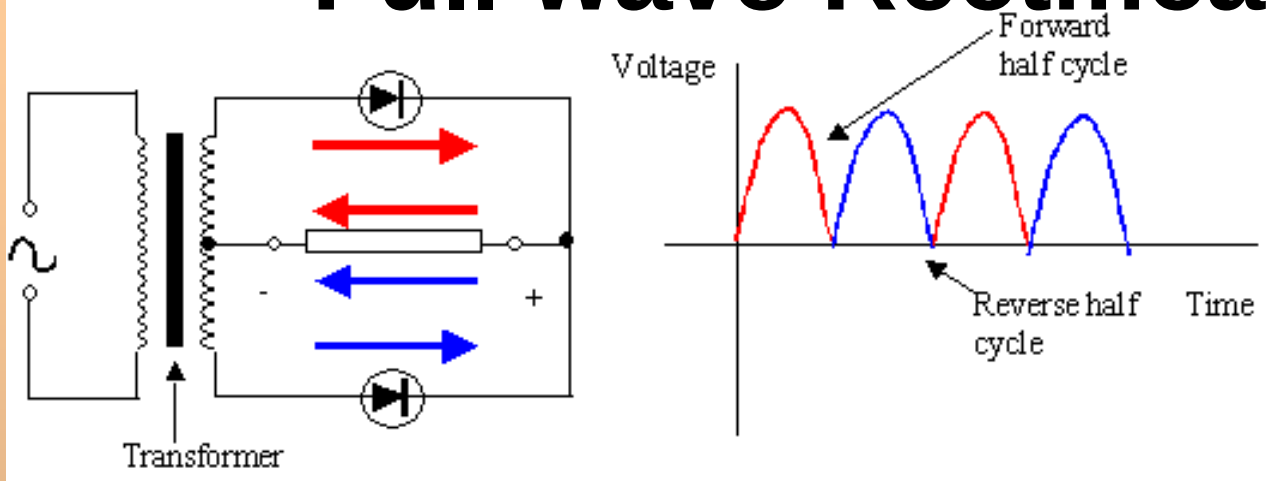
Half Wave Rectification [2]

The output is a pulsating unidirectional (direct) current. The rectifier conducts only during**half**..... the cycle.

The disadvantage is that only**half**..... cycle contributes to the rectification.

This is adequate for a crude circuit, for example the low voltage fan motor for a hair dryer.

Full wave Rectification

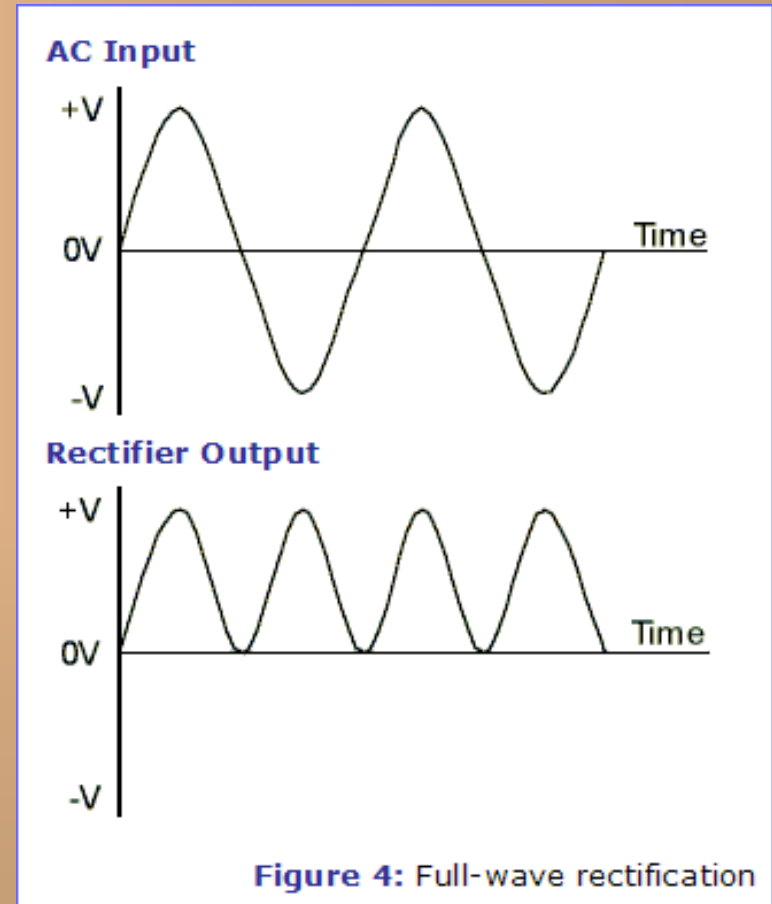
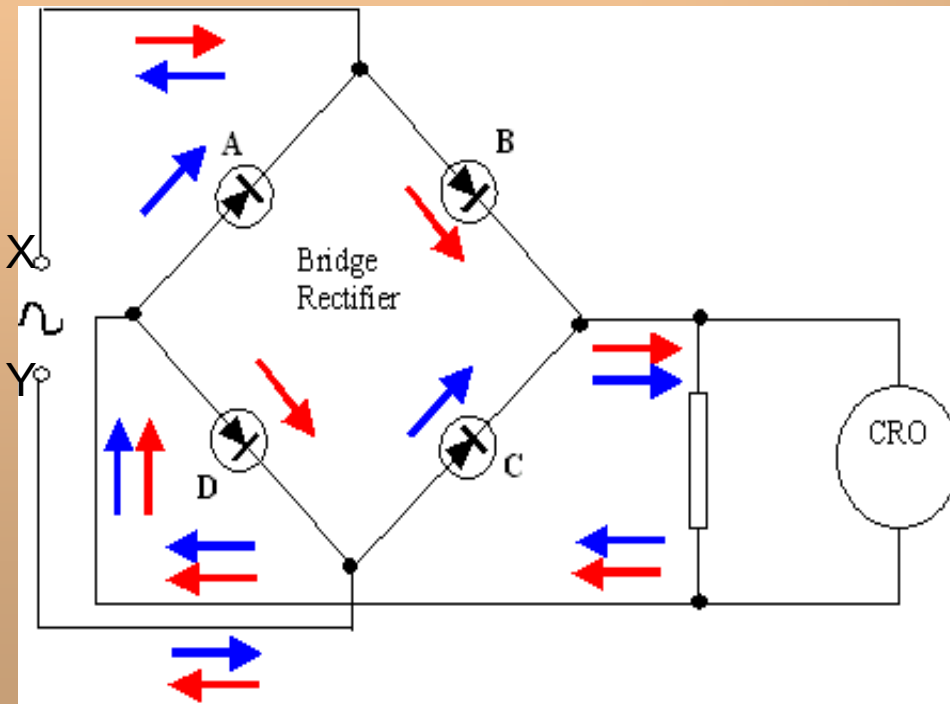


In the case above we see that both forward and reverse half cycles are rectified.

- Two half-wave rectifiers are placed back to back.
- The load is connected to a centre tapping of the transformer.
- This is called a centre-tap full-wave rectifier.
- It always needs a transformer with a centre tap.

Bridge Rectifier

The arrows show the forward and reverse half cycles:



<http://www.eleinmec.com/article.asp?18>

How it works?

During the first half cycle, when terminal X of the supply is positive, diodes **B**..... and**D** are conducting, and diodes**A**..... and**C**..... are reversed biased.

During the second half cycle, when terminal Y is positive, diodes**A**..... and**C**..... are conducting, and diodes**B**..... and**D**..... are reversed biased.

In both half cycle. the current through the load are in the**same**..... direction.

How it works?

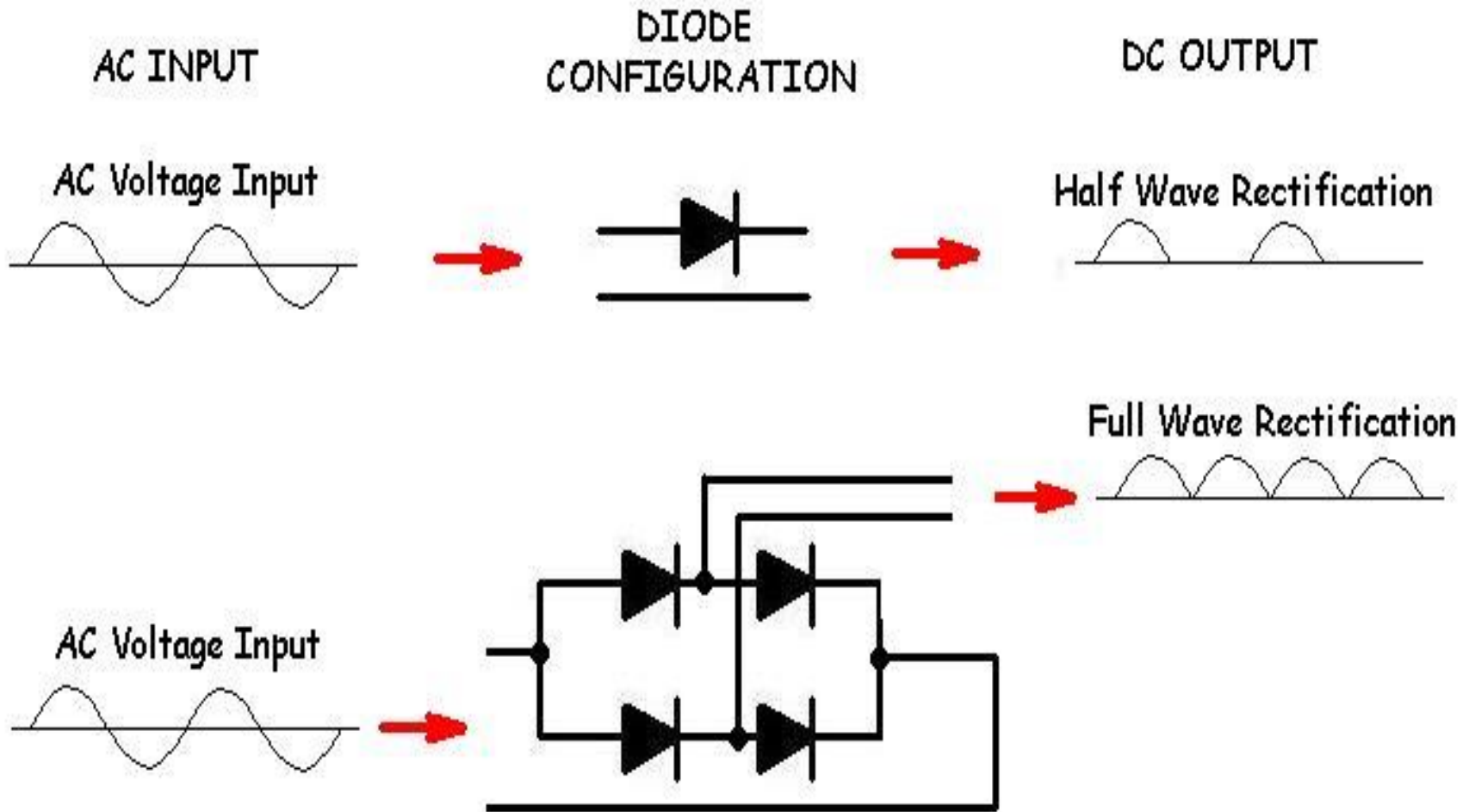
Power utilised isdoubled..... that achieved with half wave rectification.

The output isdoubled..... with an average voltage of

$$\langle V \rangle = \frac{2}{3} V_0$$

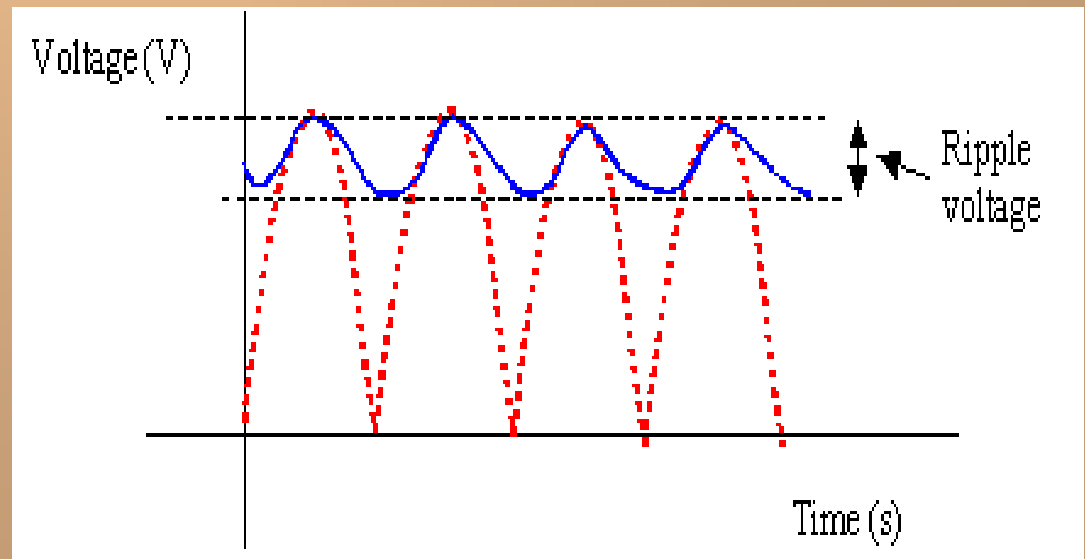
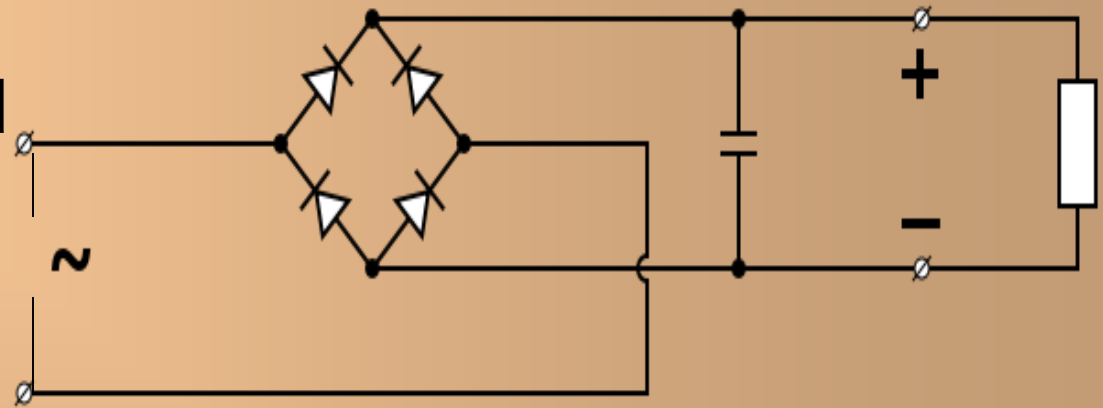
where V_0 is the peak voltage.

Alternative diagram



Smoothing

The pulsating output produced by both half-wave and full-wave rectifiers can be made more steady (smooth) by putting suitable capacitor in parallel with the load.



Smoothing [2]

When the current flows through the load in both half cycles the capacitor charges, and when the voltage across the load decreases the capacitor discharges.

If the time constant CR is large the capacitor recharges before it has completely discharged i.e. use a capacitor with **large capacity**.

The output is ripple voltage at twice the input frequency.

Summary

1. Alternating currents can be rectified using diodes;
2. A single diode will carry out half wave rectification;
3. Two diodes connected to a centre tapped transformer will carry out full wave rectification;
4. Four diodes in a bridge circuit form a bridge rectifier.
5. Capacitors are used to smooth rectified AC.